

llmXive Automated Review of Are We Ready For An Agent-Native Memory System?

llmXive automated review panel

Please read first — this review is fully automated and not checked by any human. It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

This paper's panel (9 lenses): Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

Claim Accuracy

Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.

Reviewer model: [qwen.qwen3.5-122b](#)

Prompt: `agents/prompts/paper_reviewer_claim_accuracy.md`

The paper makes several specific factual claims regarding the number of systems evaluated, the years of cited works, and the specific metric scores achieved by baselines.

First, the Introduction (Section 1) cites multiple sources with the year “2026” (e.g., `luo2026data`, `openai2026agents`, `microsoft2025copilot` where 2025 is also used). Given the paper’s arXiv ID (2606.24775) suggests a future date or a typo in the ID itself, citing works from the future as established facts is a significant accuracy issue. If these are real papers, the years must be corrected to the actual publication dates (likely 2024 or 2025). If they are internal roadmaps, they should be explicitly labeled as such rather than standard citations.

Second, there is a discrepancy in the count of evaluated systems. The Introduction states, “we evaluate 12 representative memory systems,” but Table 1 (Taxonomy) explicitly lists 13 distinct systems (MemoChat, Mem0, MEM1, MemAgent, MemTree, Zep, Mem0^g, Cognee, LightMem, SimpleMem, MemOS, MemoryOS, A-MEM). The text must be corrected to match the table, or the table must be adjusted to reflect the actual 12 systems tested.

Third, the reporting of baseline performance in Section 4.1 (RQ1) is ambiguous. The text claims “Long Context achieves 48.20 EM on DB-Bench” while simultaneously stating “MemoChat leads DB-Bench (55.40 Task Success Rate).” The metric “48.20” is explicitly labeled as EM in the text, but the context implies a comparison of overall effectiveness. If 48.20 is the EM score for Long Context, it is crucial to clarify why MemoChat is said to “lead” if its EM is lower (or if the 55.40 is a different metric). The text conflates EM and Task Success Rate without clear distinction in the narrative, potentially misleading the reader about which system is superior on which metric.

Finally, the baseline performance of “Long Context” varies across sections without explanation. In Section 4.3 (RQ3), Table 3 shows Long Context with 8.1 EM on LoCoMo. However, in Section 4.4 (RQ4), the text states Long Context drops from 42.6 to 19.0 on LongBench. While these are different datasets, the lack of a clear baseline table or explicit statement that these are different experimental conditions makes it difficult to verify the claim that “Long Context” is a consistent baseline across all evaluations. The authors should ensure that baseline performance numbers are consistently reported or clearly distinguished by dataset.

Action items.

- **[writing]** Section 1 (Introduction) claims ‘2026’ as a current year for citations (e.g., `luo2026data`, `openai2026agents`). As the paper is a preprint (arXiv:2606.24775), these future-dated citations are factually impossible unless they refer to unreleased internal roadmaps. Verify if these are typos for 2024/2025 or if the paper relies on hallucinated/future-dated sources.
- **[writing]** Section 1 claims ‘12 representative memory systems’ are evaluated, but Table 1 lists 13 distinct systems (MemoChat, Mem0, MEM1, MemAgent, MemTree, Zep, Mem0^g, Cognee, LightMem, SimpleMem, MemOS, MemoryOS, A-MEM). The count in the text does not match the evidence in the table.
- **[writing]** Section 4.1 (RQ1) states ‘Long Context achieves 48.20 EM on DB-Bench’, but the text immediately before says ‘MemoChat leads DB-Bench (55.40 Task Success Rate)’. The

claim conflates ‘Exact Match’ (EM) with ‘Task Success Rate’ (TSR) metrics, attributing the EM score to the wrong system or metric type without clarification.

- **[writing]** Section 4.3 (RQ3) Table 3 lists ‘Long Context’ with 8.1 EM on LoCoMo, but the text in Section 4.4 (RQ4) claims ‘Long Context drops (42.6 -> 19.0)’ on LongBench. The baseline performance values for ‘Long Context’ vary significantly across sections (8.1 vs 42.6) without explaining if these are different datasets or different experimental setups, creating ambiguity in the comparative claims.

Figure Critic

Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_figure_critic.md

Figure 1

Figure 1 presents four workflow diagrams with a legend that mismatches the visual encoding in the panels, uses truncated labels, and omits definitions for key components like ‘Reflection’ and ‘Planner’, reducing interpretability.

Figure 2

Figure 2 presents a scatter plot comparing memory methods but suffers from an unrefined caption and a logarithmic x-axis that lacks clear intermediate tick labels, hindering precise data interpretation.

Figure 3

The figure is a single scatter plot that contradicts the caption’s description of a two-panel layout. Additionally, the x-axis label is ambiguous regarding the aggregation of runtime data.

Figure 4

Figure 4 effectively illustrates three distinct memory representation methods (Token-Level, Graph & Tree-Based, and Heterogeneous Composite) with clear visual diagrams and labels. The internal structure of each method is well-defined, and the caption accurately reflects the content shown.

Figure 5

Figure 5 effectively illustrates three distinct memory storage architectures (Transient In-Context Register, Specialized Single-Engine, and Heterogeneous Multi-Engine) using clear diagrams and labels. The visual distinction between the ‘No External Store’, ‘Single-Specialized Store’, and ‘Multi-Distributed Store’ approaches is intuitive and aligns well with the caption ‘Memory Storage Methods’.

Figure 6

Figure 6 effectively illustrates three distinct memory extraction methods (Raw Sequence Concatenation, Schema-Free Semantic Extraction, and Schema-Constrained Structured Extraction) using clear diagrams and text examples. The visual layout is uncluttered, and the caption accurately describes the content shown.

Figure 7

Figure 7 effectively visualizes five distinct memory retrieval methods with clear, labeled diagrams and a consistent visual style. The figure is self-contained, and the caption accurately reflects the content shown.

Figure 8

Figure 8 effectively illustrates four distinct memory maintenance methods using clear diagrams and labeled sub-panels. The visual hierarchy and text are legible, and the content aligns well with the caption ‘Memory Maintenance Methods’.

Figure 9

Figure 9 presents a comprehensive comparison across multiple benchmarks, but the x-axis labels are rotated to the point of illegibility, and the legend fails to clarify how the specific methods are categorized into the four color groups.

Figure 10

Figure 10 presents retrieval results but suffers from a contradiction between the legend and bar chart labels in panel (a) regarding ‘Zep Local’ vs ‘Zep MemTree’. Additionally, the x-axis labels in the bar chart are cluttered, and the line styles in the line chart are hard to distinguish.

Figure 11

The figure displays performance metrics for various memory systems rather than a backbone ablation study as claimed in the caption, creating a significant content mismatch. Additionally, the data labels on the bars are too small to be legible.

Figure 12

Figure 12 presents three panels evaluating memory system stability, but panel (b) suffers from severe visual clutter due to overlapping lines and a dense legend, hindering interpretation. Additionally, inconsistent y-axis metrics between panels (b) and (c) and split x-axis labels in panel (a) reduce clarity.

Action items.

- **[writing]** Figure 1: The legend at the top (Memory Represent. & Storage, Memory Extraction, etc.) uses colored boxes that do not match the shapes or colors used in the diagrams below (e.g., ‘Memory Extraction’ is a green box in the legend but green hexagons in the figure). This creates ambiguity about which components belong to which category.
- **[writing]** Figure 1: The legend labels are truncated (e.g., ‘Represent. & Storage’ instead of ‘Representation & Storage’), reducing clarity and professionalism.
- **[science]** Figure 1: In panel (a), the ‘Reflection (LLM)’ and ‘Planner (LLM)’ components are gray hexagons, but they are not defined in the legend or caption — it’s unclear whether they fall under ‘Memory Retrieval / Routing’ or another category.
- **[writing]** Figure 2: The caption is marked ‘(to be refined)’ and lacks specific details about the datasets, metrics, or experimental conditions shown in the plot.
- **[science]** Figure 2: The x-axis uses a logarithmic scale (1, 3, 10, 30, 100, 300) but lacks explicit tick labels or grid alignment for intermediate values, making precise reading of runtime difficult.
- **[writing]** Figure 2: The legend ‘Double-outline = Pareto-optimal’ is present, but the term ‘Pareto-optimal’ is not defined in the caption, leaving the criteria for optimality ambiguous.
- **[fatal]** Figure 3: The caption describes a two-panel figure ((a) comparison, (b) trending), but the image shows only a single scatter plot.
- **[science]** Figure 3: The x-axis label ‘Sum of Mean Build+Query Runtime Across 3 Tasks (s)’ is ambiguous; it is unclear if the values represent the sum of means or the mean of sums.
- **[writing]** Figure 3: The legend ‘Double-outline = Pareto-optimal’ is a text box rather than a standard legend, and the term ‘Pareto-optimal’ is not defined in the caption.
- **[science]** Figure 9: The x-axis labels in subplots (a)-(d) are rotated 45 degrees, causing the text to be illegible and unreadable.
- **[science]** Figure 9: Subplots (a)-(d) lack a y-axis title (e.g., ‘Score’ or ‘Percentage’), making the metric units ambiguous despite the caption.
- **[science]** Figure 9: The legend at the top (‘Reference Baselines’, ‘Sequential Context’, etc.) does not map to the specific x-axis categories (e.g., ‘Mem0’, ‘Cognee’) shown in the plots, leaving the grouping logic unclear.
- **[science]** Figure 10: The legend lists ‘Zep Local’ (red square), but the bar chart in panel (a)

labels the corresponding red bars as ‘Zep MemTree’, creating a direct contradiction between the legend and the chart labels.

- **[science]** Figure 10: Panel (a) bar chart x-axis labels are cluttered and overlapping (e.g., ‘Zep MemTree’, ‘LightMem SimpleMem’), making it difficult to distinguish which bar belongs to which method.
- **[writing]** Figure 10: The legend in panel (b) uses inconsistent line styles (solid, dashed, dotted, dash-dot) and markers that are difficult to distinguish visually in the plot, reducing readability.
- **[science]** Figure 11: The x-axis labels (‘Emb. RAG’, ‘LightMem’, ‘MemOS’, ‘MemTree’, ‘A-MEM’) describe specific memory systems, but the caption states the figure is an ‘Ablation of Backbones’. This indicates a mismatch between the figure content and its caption, suggesting the wrong plot was rendered or the caption is incorrect.
- **[writing]** Figure 11: The numerical data labels on top of the bars are extremely small and illegible, making it impossible to read the precise values without zooming in significantly.
- **[science]** Figure 12: The legend in panel (b) lists 12 methods, but the x-axis only has 4 bins. The lines are extremely cluttered and overlapping, making it impossible to distinguish individual method trends or verify the ‘growth’ claim for specific systems.
- **[writing]** Figure 12: Panel (a) x-axis labels are split across multiple lines (e.g., ‘Long Ctx’, ‘Embed RAG’), which is visually cluttered and reduces readability compared to a single-line label.
- **[science]** Figure 12: Panel (c) y-axis is ‘Answer F1 (%)’ while panel (b) is ‘ROUGE-L F1 (%)’. The caption describes (c) as ‘Temporal evidence-distance drift’ but does not clarify if the metric change is intentional or if it should be consistent with (b) for comparison.

Jargon Police

Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_jargon_police.md

The manuscript suffers from a moderate density of unexplained jargon and acronyms that creates a barrier for readers outside the immediate sub-field of LLM retrieval optimization. While the paper aims for a “data management perspective,” it frequently relies on specific LLM implementation details (e.g., “KV-cache,” “RLGF”) and retrieval algorithm acronyms (“RRF,” “MMR”) without defining them at first use.

Specifically, in Section 3.1.1 and 3.1.2, terms like “KV-cache” and “KV tensors” are used assuming the reader knows the internal mechanics of transformer attention, which contradicts the goal of a broader systems evaluation. In Section 3.3, the use of “RRF” and “MMR”

without expansion assumes familiarity with specific fusion algorithms. Similarly, “ECL” (Section 3.2/Table 1) and “RLGF” (Section 3.4) are used as opaque labels.

The term “agent-native” in the title and throughout the text is used as a definitive category but lacks a precise definition distinguishing it from standard RAG or context engineering. Additionally, “substrates” (Section 3.1.1) is an unnecessary metaphor for “storage backends.” Finally, “Heat” scores (Section 3.4) are treated as a known metric without explaining the calculation logic in the prose. These omissions force the reader to guess the meaning of critical technical components, violating the principle of accessibility for a general data management audience.

Action items.

- **[writing]** The manuscript suffers from a moderate density of unexplained jargon and acronyms that creates a barrier for readers outside the immediate sub-field of LLM retrieval optimization. While the paper aims for a “data management perspective,” it frequently relies on specific LLM implementation details (e.g., “KV-cache,” “RLGF”) and retrieval algorithm acronyms (“RRF,” “MMR”) without defining them at first use. Specifically, in Section 3.1.1 and 3.1.2, terms like “KV-cache” and “KV tensors” are used a

Logical Consistency

Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_logical_consistency.md

The paper presents a logically coherent framework for evaluating agent memory systems, decomposing them into four distinct modules (Representation, Extraction, Retrieval, Maintenance). The causal links between architectural choices (e.g., hierarchical trees vs. flat sequences) and performance outcomes (e.g., retrieval fidelity over long horizons) are generally well-supported by the ablation studies in Section 5. The conclusion that “no single architecture dominates” follows logically from the cross-workload variance observed in Section 4.

However, there are minor logical gaps in the generalization of specific findings:

1. **Over-generalization of Graph Methods (Section 4.3):** The text asserts that “Graph-based methods handle updates reliably” as a blanket statement. While Table 2 shows Zep (a graph-based system) leading in Knowledge Update, Cognee (also graph-based) lags behind Zep in that specific metric but leads in Temporal Reasoning. The logical leap from “Zep is good at updates” to “Graph-based methods are reliable for updates” ignores the variance within the category. The mechanism (e.g., timestamp-based multi-versioning in Zep vs. triplet extraction in Cognee) should be explicitly linked to the success to support the causal claim.
2. **Contradictory Evidence in Extraction Strategy (Section 5.2):** The finding “Broader, less selective extraction preserves downstream answerability” is derived primarily from the

MemOS Fast vs. Fine comparison on LoCoMo (25.5 vs 2.5 EM). However, the same table shows “Fine Memorize” outperforming “Fast” on LongMemEval (22.3 vs 20.7 EM). The conclusion presents “broad extraction” as the superior strategy without acknowledging the context-dependency revealed by the data. The logic holds for conversational recall (LoCoMo) but fails for factual precision (LongMemEval), suggesting the causal claim requires a conditional premise regarding the workload type.

3. **Cost-Benefit Logic (Section 4.5):** The paper highlights the high latency of Cognee (116.5s) while simultaneously advocating for “localized maintenance” as the most cost-efficient rule. The text lists Cognee as a “high-utility” example but does not logically bridge the gap between its high cost and the paper’s efficiency recommendation. The conclusion implies a trade-off exists but does not explicitly state the conditions under which the high cost of global reorganization (Cognee) is justified, creating a slight disconnect between the “Operational Scaling Rule” and the specific system evaluation.

These issues do not invalidate the core findings but require precise qualification in the text to ensure the conclusions strictly follow the presented evidence.

Action items.

- **[science]** In Section 4.3, the claim that “graph-based methods handle updates reliably” overgeneralizes the data. Table 2 shows Zep leads in Knowledge Update, but Cognee (also graph-based) lags there while leading in Temporal Reasoning. The causal link between “graph-based” and “reliable updates” is not uniform; specify the mechanism (e.g., multi-versioning) driving Zep’s success.
- **[writing]** In Section 5.2, the conclusion that “broad extraction preserves answerability” ignores that MemOS Fine outperforms Fast on LongMemEval (22.3 vs 20.7 EM). The claim is only supported for LoCoMo. Qualify the conclusion to reflect that broad extraction is superior for conversational recall but selective extraction may be better for factual precision.
- **[writing]** In Section 4.5, the paper cites Cognee’s high latency (116.5s) as a “high-utility” example while advocating for “localized maintenance” as the most cost-efficient rule. The text fails to logically reconcile why a high-cost system is highlighted as a positive example without explicitly stating the specific workload conditions where such cost is justified.

Overreach

Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_overreach.md

The paper exhibits significant overreach in its framing and specific claims relative to the presented data.

First, the title “Are We Ready For An Agent-Native Memory System?” and the abstract’s

promise to determine readiness imply a binary or definitive conclusion about the state of the field. However, the core finding (Introduction, Section 4.1) is that “no single architecture dominates” and performance is strictly “workload-dependent.” The paper fails to define a quantitative threshold for “readiness” (e.g., a specific latency/accuracy trade-off required for production) or to conclude that the field is *not* ready. Instead, it offers a nuanced taxonomy and trade-off analysis, making the definitive “readiness” question in the title an over-claim that the data does not support.

Second, there is a factual discrepancy regarding the scope of the evaluation. The abstract explicitly states, “we evaluate 12 representative memory systems.” However, Table 1 (Taxonomy) lists 13 distinct systems: MemoChat, Mem0, MEM1, MemAgent, MemTree, Zep, Mem0^g, Cognee, LightMem, SimpleMem, MemOS, MemoryOS, and A-MEM. Unless one of these was excluded from the final quantitative evaluation without explanation, the claim of evaluating “12” systems is inaccurate. If one was excluded, the abstract over-claims the comprehensiveness of the study by not specifying the exclusion criteria.

Third, the claim in the Introduction that “Raw long-context retrieval outperforms memory-backed approaches for time-dependent queries” is an over-generalization. While Section 4.1 (O1) notes that “Long Context” achieves a high Exact Match (48.20) on DB-Bench, it simultaneously reports that “MemoChat” (a memory-backed system) achieves a higher Task Success Rate (55.40) on the same benchmark. The paper conflates “Exact Match” with overall effectiveness, ignoring that the memory system outperformed the raw context baseline on the more complex Task Success metric. This selective interpretation overstates the superiority of raw retrieval.

Finally, the conclusion states that “Testbed and framework to be released,” yet the abstract and conclusion also claim “Code is available at [URL].” This is a minor logical inconsistency, but the primary overreach remains the mismatch between the definitive “readiness” framing and the conditional, workload-dependent results.

Action items.

- **[writing]** The title asks ‘Are We Ready?’, implying a definitive readiness assessment, but the conclusion states ‘no single architecture dominates’ and effectiveness is ‘workload-dependent’. The paper over-claims a readiness verdict without providing a clear, quantitative threshold for ‘readiness’ or a specific recommendation for production deployment.
- **[writing]** The abstract claims to evaluate ‘12 representative memory systems’, yet Table 1 lists 13 distinct systems (MemoChat, Mem0, MEM1, MemAgent, MemTree, Zep, Mem0^g, Cognee, LightMem, SimpleMem, MemOS, MemoryOS, A-MEM). This discrepancy suggests either an unexplained exclusion or an over-claim of the evaluation scope.
- **[science]** The paper claims ‘Raw long-context retrieval outperforms memory-backed approaches for time-dependent queries’ (Introduction). However, Section 4.1 (O1) shows ‘Long Context’ achieving 48.20 EM on DB-Bench, while ‘MemoChat’ (a memory system) achieves 55.40 Task Success Rate. The claim of raw retrieval superiority is not universally supported by the presented metrics and over-generalizes the findings.

Safety Ethics

Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_safety_ethics.md

The paper presents a comprehensive technical evaluation of agent memory systems, focusing on architectural trade-offs and performance metrics. From a safety and ethics perspective, the manuscript is generally sound as it is a benchmarking study rather than a deployment of a new dual-use capability. However, there are specific areas regarding data privacy and consent that require clarification before publication.

First, the methodology relies heavily on LLM-driven extraction and consolidation (Section 2.2 and 2.4). Systems like Mem0 and Zep extract “discrete facts” or “triplets” from input streams. The paper does not explicitly state whether the evaluation datasets (e.g., LoCoMo, LongMemEval) contain Personally Identifiable Information (PII) or sensitive user data. If the datasets contain real user conversations, the authors must confirm that appropriate redaction or anonymization was performed prior to the extraction phase to prevent the leakage of private information into the memory stores or the evaluation logs. A statement in Section 3 (Method Overview) or the Data Availability section regarding PII handling is necessary.

Second, while the paper cites external repositories for code and data (e.g., GitHub links in the Abstract and Conclusion), it does not explicitly discuss the licensing or ethical provenance of the benchmark datasets used for the “dynamic updates” and “temporal reasoning” experiments. Authors should verify and state that the datasets used for training or evaluating the memory systems do not violate copyright or contain non-consensual personal data, ensuring the research adheres to standard data ethics guidelines.

Finally, the discussion on “Memory Maintenance” (Section 2.4) covers automated eviction and semantic consolidation. While this is a technical evaluation, the authors should briefly acknowledge the ethical implications of automated “forgetting” mechanisms. In a real-world deployment, aggressive eviction or summarization could lead to the loss of critical user history or evidence. A brief sentence in the Conclusion or Discussion regarding the need for user-controlled retention policies would strengthen the ethical framing of the work.

No fatal safety risks (such as dual-use for surveillance or generating harmful content) were identified, as the paper focuses on system architecture and retrieval fidelity. The requested revisions are primarily to ensure transparency regarding data privacy and consent.

Action items.

- **[writing]** The paper evaluates 12 systems using LLMs for extraction and consolidation (e.g., Mem0, Zep, Cognee) but lacks an explicit discussion of privacy risks. Authors must address how PII (Personally Identifiable Information) is handled during the ‘Schema-Constrained Extraction’ phase, specifically whether data is redacted before storage or if the evaluation datasets contain sensitive user data that was not consented for this specific benchmarking.
- **[writing]** The evaluation relies on external datasets (e.g., LoCoMo, LongMemEval) and code

hosted on GitHub. While external hosting is acceptable, the paper must explicitly state the data licensing terms and confirm that the datasets used for ‘dynamic updates’ and ‘temporal reasoning’ do not contain non-consensual personal data or copyrighted material that would violate ethical guidelines for redistribution or automated processing.

- **[writing]** The ‘Memory Maintenance’ section discusses ‘Capacity-Driven Physical Eviction’ and ‘Semantic Consolidation.’ The authors should briefly address the ethical implications of automated data deletion or summarization, particularly if these systems were deployed in real-world scenarios where ‘forgetting’ could lead to the loss of critical evidence or the erasure of user history without explicit user control.

Scientific Evidence

Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_scientific_evidence.md`

The paper presents a comprehensive empirical study of agent memory systems, decomposing them into four modules and evaluating 12 systems across multiple benchmarks. While the scope is ambitious and the taxonomy is well-structured, the scientific evidence supporting the central claims suffers from a lack of statistical rigor and transparency in experimental reporting.

First, the manuscript consistently reports point estimates (e.g., “Zep leads LongMemEval (48.0 Judge Accuracy)” in Section 5.1) without providing measures of variance such as standard deviations, confidence intervals, or p-values. In the absence of statistical significance testing, it is impossible to determine whether the reported differences between systems (e.g., 48.0 vs. 48.20 for Long Context) are genuine performance gaps or artifacts of random variation. This is particularly critical in Section 5.1 where the claim “No single system dominates” relies on small margins that may not be statistically significant.

Second, the experimental design lacks transparency regarding sample sizes and replication. While the paper mentions evaluating systems across “11 datasets,” it does not provide a clear mapping of which systems were tested on which datasets, nor does it specify the number of random seeds used for each experiment. In Section 6 (Fine-Grained Component Comparison), the ablation studies (e.g., Table 4, Table 5) present results for specific variants (e.g., “User-Only Raw” vs. “User-Only Summary”) but fail to state whether these results are averages over multiple runs or single-shot measurements. Without this information, the robustness of the conclusions regarding “Conservative Consolidation” or “Late Filtering” cannot be verified.

Third, the baseline comparisons are ambiguous. The “Long Context” baseline is frequently used as a reference point (e.g., Table 3, Figure 2), but the paper does not clarify if this represents a specific model configuration (e.g., a 128k context window LLM) or an aggregate of different models. Furthermore, the lack of variance metrics for this baseline makes it difficult to assess the reliability of the claim that “Raw long-context retrieval outperforms memory-backed approaches for time-dependent queries.”

Finally, the cost analysis in Section 5.5 (RQ5) reports latency and utility metrics but does not account for the variance in these measurements. Given that latency can fluctuate significantly due to system load or network conditions, reporting only mean values without standard deviations or confidence intervals undermines the validity of the “Operational Scaling Rule” conclusion.

To strengthen the scientific evidence, the authors should: (1) report standard deviations or confidence intervals for all quantitative results; (2) perform statistical significance tests (e.g., t-tests or ANOVA) to validate performance differences; (3) explicitly state the number of random seeds and runs used for each experiment; and (4) provide a clear appendix detailing the dataset-to-system mapping and baseline configurations.

Action items.

- **[science]** The paper claims to evaluate 12 systems across 11 datasets but lacks a dedicated table or appendix listing the specific dataset-to-system mapping. Without this, the sample size and coverage of the experimental design cannot be independently verified.
- **[science]** Statistical significance testing is absent. The paper reports point estimates (e.g., ‘Zep leads... 48.0’) but provides no p-values, confidence intervals, or standard deviations to determine if observed differences (e.g., 48.0 vs 48.20) are statistically meaningful or noise.
- **[science]** The ablation studies (Section 6) modify components of specific systems (e.g., LightMem, MemOS) but do not explicitly state the number of random seeds or runs averaged for each reported metric. This omission prevents assessment of result stability and variance.
- **[science]** The ‘Long Context’ baseline is treated as a single data point in several comparisons (e.g., Table 3, Section 5.1) without clarifying if it represents a specific model configuration or an aggregate. The lack of variance metrics for this critical baseline weakens the comparative claims.

Statistical Analysis

Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.

Reviewer model: `qwen.qwen3.5-122b`

Prompt: `agents/prompts/paper_reviewer_statistical_analysis.md`

The paper presents a comprehensive empirical study of agent memory systems, decomposing them into four modules and evaluating 12 systems across 11 datasets. However, the statistical rigor of the analysis requires strengthening to support the definitive claims made about architectural superiority and trade-offs.

First, the reporting of results relies almost exclusively on point estimates (e.g., “Zep leads... 44.4 Substring EM” in Table 1, Section 4.3; “LightMem... 48.3 Utility” in Section 4.5). There is no mention of standard deviations, confidence intervals, or variance across runs. Given that LLM-based evaluations are inherently stochastic and the number of systems is relatively small ($n=12$), the absence of variance metrics makes it impossible to determine if the observed differences (e.g., between Zep and Cognee in Table 1) are statistically significant or within the margin of error.

The authors should re-run experiments to collect multiple seeds or report variance metrics to substantiate claims of “leading” performance.

Second, the ablation studies in Section 5 (Tables 2, 3, and 4) involve numerous pairwise comparisons across different system variants. The text highlights specific “winning” configurations (e.g., “Planning Only” beats “No Planning” in Table 3) without addressing the multiple-comparisons problem. Without correction (e.g., Bonferroni or Holm) or a discussion of the increased false discovery rate, these specific claims of superiority are statistically fragile.

Finally, the cost analysis in Section 4.5 (RQ5) presents latency figures (e.g., 3.67s vs 116.5s) as definitive operational costs. Latency in distributed or LLM-driven systems often exhibits high variance (heavy-tailed distributions). Reporting only the mean or a single “average” value without standard deviation or outlier analysis (beyond the mention of “outlier-filtered” in the text) obscures the reliability of these systems in production. The authors should provide distributional statistics (e.g., 95% confidence intervals) for all latency and utility metrics to ensure the cost-performance trade-offs are robust.

Action items.

- **[science]** Section 4.2 (RQ2) and Table 1 (RQ3) report point estimates (e.g., 44.4, 36.8) without confidence intervals or standard deviations. Given the stochastic nature of LLM backbones and the small sample of systems ($n=12$), statistical significance of the reported differences (e.g., Zep vs. Cognee) cannot be assessed. Please report variance metrics or perform significance testing (e.g., bootstrap or t-tests) to support claims of superiority.
- **[science]** Section 4.5 (RQ5) and Section 5.4 (M4) discuss cost-performance trade-offs and maintenance strategies based on single-run latency/accuracy points. The analysis lacks error bars or variance analysis for latency measurements, which are critical for distinguishing between system noise and genuine architectural differences. Please include standard deviations or confidence intervals for all latency and utility metrics.
- **[science]** The ablation studies in Section 5 (Tables 2-4) modify single components but do not explicitly address multiple-comparisons correction. With numerous pairwise comparisons across 12 systems and multiple variants, the risk of Type I errors is elevated. Please clarify if corrections (e.g., Bonferroni, Holm) were applied or discuss this limitation in the context of the reported ‘best’ variants.

Writing Quality

Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.

Reviewer model: qwen.qwen3.5-122b

Prompt: agents/prompts/paper_reviewer_writing_quality.md

The manuscript presents a comprehensive survey and evaluation of agent memory systems, but the writing quality is currently compromised by significant structural inconsistencies in the LaTeX source that hinder readability and logical flow.

The most critical issue is the presence of duplicate and conflicting section definitions. The “Preliminaries” section (Section 2) appears in two distinct chunks (e000 and e002) with different text, table formatting, and content depth. Similarly, the “Memory Retrieval and Query Routing” subsection is defined twice (e001 and e003) with divergent descriptions. This duplication suggests a compilation error in the source file that must be resolved to ensure the paper reads as a single, coherent narrative.

Furthermore, the internal numbering of observations (O1, O2, etc.) is inconsistent across the “End-to-End Assessment” section. While Section 3.2 introduces O1 and O2, Section 3.3 jumps to O4 and O5, and Section 3.4 introduces O6. This disjointed numbering confuses the reader regarding the progression of findings. A global re-numbering of observations (O1 through O11) or a clear per-subsection reset is required.

Finally, the formatting of the “Contributions” list in the Introduction lacks uniformity. The first two items follow a “Bold Header (Section Ref)” pattern, whereas the latter two switch to “Bold Header: **Bold Question**“. Standardizing this list will improve the professional polish of the introduction.

Addressing these structural and formatting issues is essential before the paper can be considered for acceptance.

Action items.

- **[writing]** The LaTeX source contains duplicate section definitions. Section ‘Preliminaries’ (sec:def) appears in both e000 and e002 with conflicting content and table formatting. Section ‘Memory Retrieval and Query Routing’ (subsec:retrieval) is defined twice (in e001 and e003) with different text. This causes compilation errors and narrative confusion. Merge these sections into single, coherent definitions.
- **[writing]** In Section 1 (Introduction), the list of contributions uses inconsistent formatting. Items (1) and (2) use bold headers followed by parenthetical section references, while items (3) and (4) use bold headers followed by bolded questions. Standardize the structure for all four contributions to improve readability.
- **[writing]** Section 3.2 (Memory Retrieval Fidelity) and Section 3.5 (Memory Operation Cost) contain ‘O1’ and ‘O7’ labels respectively, but Section 3.3 (Memory Evolution Robustness) jumps to ‘O4’ and ‘O5’, and Section 3.4 (Long Horizon Memory Stability) uses ‘O6’. The observation numbering is non-sequential and disjointed across subsections. Re-number observations globally (O1-O11) or per subsection to ensure logical flow.